

EXPERIMENT 23: Prolog Program to Implement Family Tree**Aim**

To write and execute a Prolog program to represent a family tree and determine relationships such as parent, child, sibling, and grandparent using facts and rules.

Algorithm

1. Start the Prolog program.
2. Define facts representing parent-child relationships.
3. Define rules for parent, child, sibling, and grandparent relationships.
4. Use logical rules to derive indirect relationships.
5. Execute queries for the required family relationship.
6. Display the result.
7. Stop.

Program Code:

% Family tree

parent(john, sam).

parent(john, mary).

parent(susan, sam).

parent(susan, mary).

parent(sam, alex).

parent(sam, lisa).

parent(mary, david).

% Father

father(X, Y) :-

```
parent(X, Y),  
male(X).
```

```
% Mother  
mother(X, Y) :-  
    parent(X, Y),  
    female(X).
```

```
% Gender  
male(john).  
male(sam).  
male(alex).  
male(david).
```

```
female(susan).  
female(mary).  
female(lisa).
```

```
% Sibling  
sibling(X, Y) :-  
    parent(P, X),  
    parent(P, Y),  
    X \= Y.
```

```
% Grandparent  
grandparent(X, Y) :-  
    parent(X, Z),
```

```
parent(Z, Y).
```

Output:

```
(109 ms) yes
| ?- consult('C:/Users/Hp/Documents/ex-23.pl').
compiling C:/Users/Hp/Documents/ex-23.pl for byte code...
C:/Users/Hp/Documents/ex-23.pl compiled, 41 lines read - 2593 bytes written, 39 ms

yes
| ?- sibling(sam, mary).

true ? g
Action (; for next solution, a for all solutions, RET to stop) ? r
Action (; for next solution, a for all solutions, RET to stop) ? a

true

(16 ms) no
| ?- ndparent(john, alex).|
```

Result

The Prolog program for implementing the family tree was successfully developed and executed. The required family relationships were obtained correctly using Prolog facts, rules, and queries.